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Vehicle wheel spoke connection

Abstract

A vehicle wheel having a rim, a hub, a plurality of spokes extending between said rim and said hub, rim and/or hub includes a spoke bed wall with an outboard bed surface, an inboard bed surface, and a spoke hole therethrough having a longitudinally inboard entrance. The spoke includes: a first spoke portion and a second spoke portion longitudinally inboard of the first spoke portion and laterally outboard of the first spoke portion, including a transition surface therebetween. The spoke extends through the spoke hole and at least a portion of the transition surface is longitudinally outboard of the inboard entrance and the spoke is connected to the rim and/or hub at the first spoke portion.

Inventors:	Schlanger; Raphael (Wilton, CT)
Applicant:	Schlanger; Raphael (Wilton, CT)
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Primary Examiner: Morano; S. Joseph

Assistant Examiner: Castonguay; Emily G.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority of Provisional Patent Application Ser. No. 63/186,133, filed May 9, 2021; (2) This application is also a Continuation-In-Part of U.S. patent application Ser. No. 16/988,666, filed Aug. 9, 2020 and currently pending; (3) U.S. patent application Ser. No. 16/988,666 claims priority of Provisional Patent Application Ser. No. 62/884,684, filed Aug. 9, 2019; (4) U.S. patent application Ser. No. 16/988,666 is also a Continuation-In-Part of U.S. patent application Ser. No. 17/020,852, filed Sep. 15, 2020 and currently pending; (5) U.S. patent application Ser. No. 17/020,852 is a Continuation-In-Part of U.S. patent application Ser. No. 15/412,151, filed Jan. 23, 2017 and as U.S. Pat. No. 10,814,671 on Oct. 27, 2020; (6) U.S. patent application Ser. No. 15/412,151 claims priority of Provisional Patent Application Ser. No. 62/289,389, filed Feb. 1, 2016; (7) U.S. patent application Ser. No. 15/412,151 is also a Continuation-In-Part of U.S. patent application Ser. No. 14/641,615, filed Mar. 9, 2015 and issued as U.S. Pat. No. 9,815,321 on Nov. 14, 2017; (8) U.S. patent application Ser. No. 14/641,615 is a Continuation-In-Part of U.S. patent application Ser. No. 12/930,643, filed Jan. 13, 2011 and issued as U.S. Pat. No. 8,985,709 on Mar. 24, 2015; (9) U.S. patent Ser. No. 12/930,643 is a Continuation-In-Part of U.S. patent application Ser. No. 12/806,064, filed Aug. 5, 2010 and issued as U.S. Pat. No. 8,657,387 on Feb. 25, 2014; (10) U.S. patent application Ser. No. 12/806,064 is a Continuation-In-Part of U.S. patent application Ser. No. 11/879,333, filed Jul. 17, 2007 and issued as U.S. Pat. No. 7,784,878 on Aug. 31, 2010; (11) U.S.

patent application Ser. No. 11/879,333 is a Continuation-In-Part of U.S. patent application Ser. No. 10/755,653, filed Jan. 12, 2004 and issued as U.S. Pat. No. 7,357,460 on Apr. 15, 2008; (12) U.S. patent application Ser. No. 10/755,653 claims priority of U.S. provisional patent application Ser. No. 60/439,819, filed Jan. 13, 2003.

BACKGROUND OF THE INVENTION

- (1) The present invention relates to spoked vehicle wheels and bicycle wheels in particular. More specifically, this invention relates to the interface between the spoke and the spoke bed of the bracing element. The bracing element constitutes the rim and/or hub of the vehicle wheel.
- (2) Prior art wheels commonly include wire spokes that are connected to the rim by mating spoke nipples. These nipples are commonly termed “external nipples” if the nipple passes radially inwardly through the spoke hole of the rim so that the nipple is exposed and may be manipulated at a location radially inboard of the rim's spoke bed wall. The nipples are commonly termed “internal nipples” if the nipple does not extend radially inboard of the spoke bed wall and if the nipple may be manipulated through access at a point radially outboard of the spoke bed wall.
- (3) Particularly with the advent of spokes made of carbon fiber composite or other alternative materials, the spoke may be configured to have a laterally narrower first portion thereof that is positioned longitudinally outwardly from a laterally wider portion thereof. For example, the first portion may be configured to mimic a traditional wire spoke having external threads, while the second portion may be of carbon fiber composite or other alternate material.
- (4) In a conventional spoke/nipple/rim arrangement, the nipple may be threadably assembled to the narrower first portion of the spoke. This leaves a portion of the first portion exposed to allow for the requisite threadable adjustment. Thus the spoke is necked down to expose the first portion, resulting in an unsightly transition between the spoke and the nipple and/or the rim. The result is a wheel that has poor aesthetics and is visually objectionable to the user.
- (5) In addition, the stepped transition between the first portion, the second portion, the nipple, and/or the rim creates multiple steps in geometry that is disruptive to air flow in the aerodynamically-critical region between the spoke and the rim. The potential result is greater aerodynamic drag due to this increased number of steps in geometry.
- (6) It is a goal of the present invention to create a spoke connection that is visually clean and has good aesthetics. It is a further goal to optimize aerodynamics of the wheel by creating a smoother transition between the spoke and the rim and/or hub, with fewer exposed steps in geometry.

SUMMARY OF THE INVENTION

- (7) The present invention includes a spoke that has a longitudinally outward portion of smaller lateral dimension than a longitudinally inward portion thereof. This creates a laterally extending transition step between the longitudinally outward and longitudinally inward portions. The spoke passes through an opening in the rim and/or hub and the transition step is positioned to be longitudinally outward of the longitudinally inward entrance to the opening.
- (8) The embodiments of the present invention are focused on shielding the transition step between the first and second portions of the spoke. By visually shielding this transition, the aesthetics of the wheel is improved. Furthermore, by shielding this transition, the number of aforementioned steps in geometry that are exposed to the wind is reduced, resulting in a wheel with enhanced aerodynamics.
- (9) Further features of the present invention will become apparent from considering the drawings and ensuing description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention will be more readily understandable from a consideration of the

accompanying exemplificative drawings, wherein:

- (2) FIG. 1a is a perspective view schematically illustrating the general configuration of a vehicle wheel as applied to a bicycle wheel.
- (3) FIG. 1B is an axial plane cross-sectional view of the wheel of FIG. 1a, taken along 3-3;
- (4) FIG. 2a is a partial perspective exploded view of a wheel of prior art design, including an "external" spoke nipple, with the rim shown in axial plane cross-section;
- (5) FIG. 2b is an enlarged detail of a portion of the rim of FIG. 2a, with the rim shown in axial plane cross-section;
- (6) FIG. 2c is a partial perspective view, with the rim in axial plane cross-section, showing the wheel of FIG. 2a with the spoke and nipple assembled to the rim;
- (7) FIG. 2d is an axial plane cross-section detail view of the wheel of FIG. 2a in the configuration shown in FIG. 2c;
- (8) FIG. 2e is an axial plane cross-section detail view corresponding to boundary 57 of FIG. 2d;
- (9) FIG. 3a is a partial perspective exploded detail view of a wheel of prior art design, including an "internal" spoke nipple, with the rim shown in axial plane cross-section;
- (10) FIG. 3b is a partial perspective detail view, with the rim in axial plane cross-section, showing the wheel of FIG. 3a with the spoke and nipple assembled to the rim;
- (11) FIG. 3c is an axial plane cross-section detail view of the wheel of FIG. 3a, corresponding to the configuration shown in FIG. 3a;
- (12) FIG. 3d is an axial plane cross-section detail view of the wheel of FIG. 3a, corresponding to the configuration shown in FIG. 3b;
- (13) FIG. 3e is an orthogonal detail view, taken along 67-67, showing the spoke of FIG. 3a;
- (14) FIG. 3f is an orthogonal detail view, taken along 68-68, showing the spoke of FIG. 3a;
- (15) FIG. 3g is perspective detail view, showing the spoke of FIG. 3a;
- (16) FIG. 4a is a partial perspective exploded detail view of a first embodiment of the present invention, with the rim shown in axial plane cross-section;
- (17) FIG. 4b is a partial perspective detail view, showing the embodiment of FIG. 4a, with the rim shown in axial plane cross-section, and with the spoke and spoke nipple next assembled through a spoke hole of the rim;
- (18) FIG. 4c is an axial plane cross-section exploded detail view, taken along 69-69, of the embodiment of FIG. 4a and corresponding to the assembly sequence of FIG. 4a;
- (19) FIG. 4d is an axial plane cross-section detail view, taken along 69-69, of the embodiment of FIG. 4a and corresponding to the assembly sequence of FIG. 4b, showing the enlarged portion of the spoke longitudinally overlapping the spoke hole of the rim;
- (20) FIG. 4e is an axial plane cross-section detail view, of the rim of a second embodiment of the present invention, including a stepped spoke hole;
- (21) FIG. 4f is an axial plane cross-section detail view of the embodiment of FIG. 4e and corresponding to the assembly sequence of FIG. 4d, showing the enlarged portion of the spoke longitudinally overlapping the stepped spoke hole of the rim;
- (22) FIG. 5a is a partial perspective exploded detail view of a third embodiment of the present invention, with the rim shown in axial plane cross-section;
- (23) FIG. 5b is a partial perspective detail view, showing the embodiment of FIG. 5a, with the rim shown in axial plane cross-section, and with the spoke and spoke nipple next assembled through a spoke hole of the rim;
- (24) FIG. 5c is an axial plane cross-section exploded detail view, taken along 70-70, of the embodiment of FIG. 5a and corresponding to the assembly sequence of FIG. 5a;
- (25) FIG. 5d is an axial plane cross-section detail view, taken along 70-70, of the embodiment of FIG. 5a and corresponding to the assembly sequence of FIG. 5b, showing the enlarged portion of the spoke longitudinally overlapping both the spoke hole of the rim and a recess of the spoke nipple;

- (26) FIG. 6a is a partial perspective exploded detail view of a fourth embodiment of the present invention, with the rim shown in axial plane cross-section, including an intermediate washer;
- (27) FIG. 6b is a partial perspective detail view, showing the embodiment of FIG. 6a, with the rim shown in axial plane cross-section, and with the spoke, intermediate washer, and spoke nipple next assembled through a spoke hole of the rim;
- (28) FIG. 6c is an axial plane cross-section exploded detail view, taken along 71-71, of the embodiment of FIG. 6a and corresponding to the assembly sequence of FIG. 6a;
- (29) FIG. 6d is an axial plane cross-section detail view, taken along 71-71, of the embodiment of FIG. 6a and corresponding to the assembly sequence of FIG. 6b, showing the enlarged portion of the spoke longitudinally overlapping both the spoke hole of the rim and a recess of the intermediate washer;
- (30) FIG. 7a is a partial perspective exploded detail view of a fifth embodiment of the present invention, with the rim shown in axial plane cross-section, including an intermediate sleeve;
- (31) FIG. 7b is a partial perspective detail view, showing the embodiment of FIG. 7a, with the rim shown in axial plane cross-section, and with the spoke, intermediate sleeve, and spoke nipple next assembled through a spoke hole of the rim;
- (32) FIG. 7c is an axial plane cross-section exploded detail view, taken along 72-72, of the embodiment of FIG. 7a and corresponding to the assembly sequence of FIG. 7a;
- (33) FIG. 7d is an axial plane cross-section detail view, taken along 72-72, of the embodiment of FIG. 7a and corresponding to the assembly sequence of FIG. 7b, showing the enlarged portion of the spoke longitudinally overlapping both the spoke hole of the rim and the opening of the intermediate sleeve;
- (34) FIG. 8a is a partial perspective exploded detail view of a sixth embodiment of the present invention, with the rim shown in axial plane cross-section, including a stepped sleeve;
- (35) FIG. 8b is a partial perspective detail view, showing the embodiment of FIG. 8a, with the rim shown in axial plane cross-section, and with the spoke, stepped sleeve, and spoke nipple next assembled through a spoke hole of the rim;
- (36) FIG. 8c is an axial plane cross-section exploded detail view, taken along 73-73, of the embodiment of FIG. 8a and corresponding to the assembly sequence of FIG. 8a;
- (37) FIG. 8d is an axial plane cross-section detail view, taken along 73-73, of the embodiment of FIG. 8a and corresponding to the assembly sequence of FIG. 8b, showing the enlarged portion of the spoke longitudinally overlapping both the spoke hole of the rim and the opening of the stepped sleeve, and showing the stepped sleeve longitudinally overlapping the spoke hole.
- (38) FIG. 9a is an axial plane cross-section exploded view of a seventh embodiment of the present invention, corresponding to the assembly sequence of FIG. 4c, showing a nipple having a pilot collar to maintain laterally piloted alignment between the nipple and the spoke hole of the rim.
- (39) FIG. 9b is an axial plane cross-section detail view of the embodiment of FIG. 9a, corresponding to the assembly sequence of FIG. 4d, showing the pilot collar longitudinally overlapping the spoke hole.

DETAILED DESCRIPTION OF THE INVENTION

(40) FIGS. 1a-b describe the basic configuration of an exemplary prior art vehicle wheel, in particular, a bicycle wheel 1, as well as a description of the direction conventions used throughout this disclosure. For clarity, the bicycle frame and the quick release skewer assembly are not shown in this figure. The hub shell 14 is rotatable about the axle 9 and includes at least two axially spaced hub flanges 16a and 16b, each of which include a means for connecting with a multiplicity of spokes 2 connected thereto. Axle 9 includes end faces 11a and 11b that define the spacing of its mounting with the frame (not shown). The axial axis 28 is the axial centerline of rotation of the bicycle wheel assembly 1. The hub flanges 16a and 16b may be contiguous with the hub shell 14 or may be separately formed and assembled to the hub body 12 portion of the hub shell 14. Each spoke 2 is affixed to its respective hub flange 16a or 16b at its first end 4 and extend to attach the

rim **8** at its second ends **6**. The tire **10** is fitted to the outer periphery of the rim **8**. The wheel of FIG. **1** is generic and may be of tension-spoke or compression-spoke design.

(41) The axial direction **92** is any direction parallel with the axial axis **28**. The radial direction **93** is a direction generally perpendicular to the axial direction **92** and extending generally from the axial axis **28** radially outwardly toward the rim **8**. The tangential direction **94** is a direction generally tangent to the rim at a given radius. The circumferential direction **95** is a cylindrical vector that wraps around the axial axis **28** at a given radius. A radial plane **96** is a plane perpendicular to the axial axis **28** that extends in a generally radial direction at a given axial intercept. An axial plane **97** is a plane that is generally parallel to the axial axis. An orientation that is radially inboard (or inward) is nearer to the axial axis **28** of rotation and a radially outboard (or outward) is further from the axial axis. An axially inboard (or inward) orientation is an orientation that is axially proximal to the axial midpoint between the two end faces **11a** and **11b**. Conversely, an axially outboard (or outward) orientation is an orientation that is axially distal to the axial midpoint between the two end faces **11a** and **11b**. A radially inboard orientation is an orientation that is radially proximal to the axial axis **28** and a radially outboard orientation is an orientation that is radially distal to the axial axis **28**. An axially inwardly facing surface is a surface that faces toward the axial midpoint between the two end faces **11a** and **11b**. Conversely, an axially outwardly facing surface is a surface that faces away from the axial midpoint between the two end faces **11a** and **11b**. While it is most common for the hub shell **14** to rotate about a fixed axle **9**, there are some cases where it is desirable to permit the axle **9** to be fixed with the wheel **1** such as the case where the wheel **1** is driven by the axle **9**.

(42) For the purposes of using conventional terminology, the term “hub flange” is used herein to describe a region of the hub shell **14** to which the spokes **2** are joined. While the surface of the hub flange may be raised and flange-like in comparison to other surfaces of the hub shell **14**, this is not a requirement for the present invention and the hub flange **16** may alternatively be flush or recessed relative to other hub shell surfaces.

(43) As is well known in the art, a wheel **1** may be of tension-spoke construction, where the central hub hangs in tension by the spokes from the rim portion directly above, or it may be of compression-spoke construction, where the hub is supported by compressing the spoke directly beneath it. Since the present invention may be directed toward bicycle wheels and since the tension-spoke wheel is generally a more efficient structure than compression-spoke wheel, most of the discussion herein is focused with an eye toward tension-spoke wheel construction. However, it is anticipated that most, if not all, of the embodiments of the present invention may be adapted or otherwise applied to compression-spoke wheel construction as well. For a tension-spoke wheel, it is preferable that the wheel includes at least two hub flanges that are axially spaced on either side of the rim or, more specifically, the spoke attachment points at the rim. Thus the spokes fixed to opposite hub flanges will converge as they extend to the rim. Additionally, a tension-spoke wheel will usually be pre-tensioned during assembly to create a pre-tensioned structure of balanced spoke tension that allows the axle supporting loads to be distributed among several, if not all, of the spokes of the wheel. It is this ability to share the stresses among its spokes that helps to make the tension-spoke wheel the highly efficient structure that it is. For a compression-spoke wheel, it is often preferable to employ at least two axially spaced hub flanges, however, in the case where the spokes have sufficient bending stiffness to support the requisite lateral or side-to-side loads, only a single hub flange may be employed.

(44) The midplane **19** of the wheel **1** is a radial plane axially positioned midway between the flanges **16a** and **16b**. The spokes **2** connected to their respective flanges **16a** and **16b** commonly converge axially (by bracing angles **18a** and **18b**) to their connection point with the rim **8**.

(45) The spoke **2** is a generally long slender tensile element with a longitudinal axis **62** along its length and generally parallel to its sidewalls. The spoke **2** also has a tensile axis **61** of applied tensile load **58** that extends along the span portion of the spoke **2** between its anchor points at the

rim **8** and hub flange **16**. The tensile axis **61** is generally collinear to the longitudinal axis **62**, except where the spoke **2** is bent to deviate from the tensile axis **61**. For the purposes of definition, as relating to spokes **2** and connections thereto, the term “longitudinal” herein refers to alignment along the longitudinal axis **62**. A longitudinally inboard (or inward) orientation refers to an orientation proximal the midpoint of the span portion. Conversely, a longitudinally outboard (or outward) orientation refers to an orientation distal the midpoint of the span portion. The term “lateral” herein refers to an orientation in a direction generally perpendicular to the longitudinal axis **62**. A laterally inboard (or inward) orientation refers to an orientation proximal the longitudinal axis. Conversely, a laterally outboard (or outward) orientation refers to an orientation distal the longitudinal axis **62**.

(46) FIGS. **2a-e** describe an exemplary rim **20** of generally conventional geometry. As detailed in FIGS. **2a** and **2b**, rim **20** is of a generally hollow construction, commonly termed “double-wall” construction, and includes a radially inboard spoke bed **22** wall of thickness **23** and a radially outboard tire bed **24** wall and generally radially extending sidewalls **26a** and **26b** to define a generally hollow circumferential cavity **27**. Spoke bed **22** is a wall or web portion of the rim **20** that includes a radially inboard surface **32** and a radially outboard surface **34** and is defined herein as the portion of the rim to which the spoke is structurally connected and/or anchored. Hooked flanges **30a** and **30b** are adapted to engage the beads of a conventional tire (not shown).

(47) The spoke bed **22** is pierced with a plurality of spoke holes **36** adapted for connection with their respective spokes **2** via spoke nipples **48**. The term “nipple” and “spoke nipple” are used interchangeably herein. It may be seen that the spoke hole **36**, which extends along hole axis **35**, has a radially inboard entrance **38** at its intersection with the radially inboard surface **32** and a radially outboard entrance **40** at its intersection with the radially outboard surface **34**. Spoke hole **36** is most commonly a circular cylindrical hole that is commonly produced by drilling along hole axis **35** or by other fabrication processes. The tire bed **24** wall is pierced by access hole **37** that is preferably aligned with spoke hole **36**, to permit the nipple **48** to be assembled as shown in FIGS. **2a** and **2c**. Note that access hole **37** is merely one common means to permit the nipple **48** to be assembled to the rim; a wide range of alternative means may be substituted, including means that do not require an access hole.

(48) It is useful to understand that it is common to manufacture the rim **20** by extruding the straight profile shown here and rolling the extrusion into a circumferential hoop with its ends joined by either a welded, sleeved or pinned connection. Spoke holes **36** and access holes **37** are then drilled in their proper locations.

(49) FIG. **2a** shows an exploded view that describes the conventional arrangement by which the second end **6** of the spoke **2** is connected to the rim **20**. The second end **6** of spoke **2** includes external threads **46** to mate with internal threads **47** of spoke nipple **48**. Spoke nipple **48** includes an enlarged head **50** and a shank **52**, with a generally conical tapered transition portion **54** extending radially outwardly between the shank **52** and the underside of the head **50**. Spoke nipple **48** also includes flats **56** for engagement with a mating wrench (not shown) for manual manipulation to adjust the spoke pre-tension by adjusting the threaded engagement between external threads **46** and internal threads **47**. Nipple **48** is considered an “external” spoke nipple, since it has a shank **52** that extends through the spoke hole **36** so that its flats are exposed and may be manipulated externally to the rim **20**. FIGS. **2c-e** shows the spoke nipple **48** threadably assembled to the spoke **2** such that the transition portion **54** overlies and contacts the outboard entrance **40**. The spoke nipple **48** is thereby structurally braced against the spoke bed **22** to resist the spoke tension **58** of the spoke **2**.

(50) FIGS. **3a-g** describe a prior art arrangement that includes an “internal” or “hidden” spoke nipple **122**. As particularly shown in FIGS. **3e-g**, spoke **100** includes a fastener **102** portion having a lateral dimension **114** and having external threads **104** to threadably mate with internal threaded hole **128** of nipple **122**. Longitudinally inboard of fastener **102** is enlarged portion **106**, having a

sidewall surface **121** that is laterally outboard and offset from the corresponding sidewall surface **105** of the external threads **104** and that has lateral dimensions **116a** and **116b**. Transition surface **108** extends generally laterally between sidewall surface **105** of fastener **102** and sidewall surface **121** of enlarged portion **106** by dimensions **120a** and **120b**. Longitudinally inboard of enlarged portion **106** is stepped portion **110**, having a sidewall surface **111** that is laterally outboard and offset from the sidewall surface **121** and that has lateral dimensions **118a** and **118b**. Transition surface **112** extends generally laterally between enlarged portion **106** and stepped portion **110**. Enlarged portion **106** and stepped portion **110** are shown here to be somewhat elliptical and non-circular in cross section such that dimensions **116a** and **118a** are greater than respective dimensions **116b** and **118b**.

(51) Spoke nipple **122** includes an internally threaded hole **128** for threadable engagement with external threads **104** and a bearing surface **126** to provide an abutting overlie engagement with the outboard surface **140** on assembly as shown in FIGS. **3b** and **3d**. Bearing surface **126** has a laterally projecting diameter dimension **127** that is larger than the diameter dimension **137** of spoke hole **136** to provide an overlie engagement therebetween. Nipple **122** also includes flats **124** for manipulation with a mating wrench (not shown).

(52) Rim **130** is of a generally hollow construction, commonly termed “double-wall” construction, and includes a radially inboard spoke bed **132** wall and a radially outboard tire bed **134** wall and generally radially extending sidewalls to define a generally hollow circumferential cavity **144**. Spoke bed **132** is defined by a radially inboard surface **142** and a radially outboard surface **140**. The spoke bed **132** is pierced with a plurality of spoke holes **136** adapted for connection with their respective spokes **100** via spoke nipples **122**. Spoke holes **136** are sized with diameter dimension **137** for clearance with the fastener **102**, but are smaller than dimension **116a** of the enlarged portion **106** such that the enlarged portion **106** will not fit through. The tire bed **134** is pierced by access hole **138** that is preferably aligned with spoke hole **136**, to permit the nipple **122** to be assembled as shown in FIGS. **3b** and **3d**. Note that access hole **138** is merely one common means to permit the nipple **122** to be assembled to the rim **130**; a wide range of alternative means may be substituted, including means that do not require a access hole.

(53) As shown in FIGS. **3b** and **3d**, fastener **102** has been assembled through spoke hole **136** and the nipple **122** has been passed through access hole **138** and threadably assembled to fastener **102**, with internal threaded hole **128** threadably engaged to external threads **104**. Bearing surface **126** is abutting and bearing against outboard surface **140** in an overlie engagement therebetween to support spoke tension **58**. This is a conventional arrangement that utilizes an internal or “hidden” spoke nipple, where the nipple **122** is hidden within the cavity **144** of the rim **130** and is not exposed, particularly after a tire is mounted to the rim in a conventional arrangement. There is a longitudinal gap **146** between the transition surface **108** and the inboard surface **142**. The transition surface **108** is visually exposed and radially inwardly external to the rim **130**. In the case where the spoke **100** includes an enlarged portion **106**, the exposed portion of the thin fastener **102** within gap **146** is considered to be aesthetically unappealing and detracts from the overall visual appearance of the wheel assembly **1**. The exposed transition surface **108** further detracts from the visual aesthetics and the aerodynamic efficiency of the associated wheel assembly **1**.

(54) FIGS. **4a-d** describe an arrangement that eliminates the gap **146** and conceals the transition surface **108** to create a more visually appealing and aerodynamic wheel assembly **1**. Spoke **100** and nipple **122** are identical to that described in FIGS. **3a-g**. Rim **150** is schematically identical to rim **130** of FIGS. **3a-d** and includes a radially inboard spoke bed **152** wall and a radially outboard tire bed **154** wall and generally radially extending sidewalls to define a generally hollow circumferential cavity **164**. Spoke bed **152** is a wall defined by a radially inboard surface **162** and a radially outboard surface **160**, including thickness **166** therebetween. It is noted that outboard surface is a curved convex surface as shown here to provide enhanced alignment with the nipple **122** upon assembly. In this example, the radially inboard periphery **153** of the rim **150** includes the

inboard surface **162**. The spoke bed **152** is pierced with a plurality of spoke holes **156** adapted for connection with their respective spokes **100** via spoke nipples **122**. Spoke holes **156** are sized with dimension **157** that is larger than that of spoke holes **136** of FIGS. **3a-d** and are sized to receive the enlarged portion **106** (as described in FIGS. **4b** and **4d**). Spoke holes **156** have a longitudinally inboard entrance **155** at its intersection with the inboard surface **162** and a longitudinally outboard entrance **159**. Spoke holes **156** are shown to be circular cylindrical openings such that they may be easily produced by drilling, as is the common fabrication method. However, the spoke hole may alternatively have a non-circular or other profile as may be preferred for functional or fabrication purposes. Spoke bed thickness **166** roughly corresponds to the longitudinal length of the spoke hole **156**. Diameter dimension **127** of bearing face **126** is larger than diameter dimension **157** of spoke hole **156** for an overlaid engagement therebetween upon assembly.

(55) The spoke **100** includes an optional stepped portion **110** and the spoke hole **156** is shown here to be optionally smaller than dimension **118a** of the stepped portion **110** such that the stepped portion **110** will not fit through and to minimize lateral clearance between the enlarged portion **106** and the spoke hole **156**. The tire bed **154** is pierced by access hole **158** that is preferably aligned with spoke hole **156**, to permit the nipple **122** to be assembled as shown in FIGS. **4b** and **4d**. Note that access hole **158** is merely one common means to permit the nipple **122** to be assembled to the rim **150**.

(56) As shown in FIGS. **4b** and **4d**, fastener **102** has been assembled through spoke hole **156** while the nipple **122** has been passed through access hole **138** and threadably assembled to fastener **102**, with internal threaded hole **128** threadably engaged to external threads **104**. Bearing surface **126** is abutting and bearing against outboard surface **160** in a structural overlaid engagement therebetween to support spoke tension **58** and to connect the spoke **100** to the rim **150**. It is noted that, in comparison to the arrangement of FIGS. **3a-d**, FIGS. **4b** and **4d** show the spoke **100** as advanced longitudinally outwardly such that the transition surface **108** is longitudinally outboard of the entrance **155** by dimension **168** such that both the transition surface **108** and the enlarged portion **106** are also longitudinally overlapping the spoke hole **156**.

(57) FIGS. **4b** and **4d** show the transition surface **108** as positioned to be fully overlapped within the spoke hole **156**. However, it is noted that outboard surface **160** and inboard surface **162** may have arcuate contours as shown such that corresponding entrances **159** and **155** are saddle-shaped and radially variable around their perimeter. Further, the spoke **100** is shown to be inclined corresponding to a bracing angle. These geometric arrangements may alternatively permit that, if the transition surface **108** is close to entrance **155**, it is envisioned that only a portion of the transition surface **108** could be longitudinally overlapping the spoke hole **156**, with the remainder of the transition surface **108** exposed and longitudinally inboard of the spoke hole **156**.

(58) The spoke hole **156** circumscribes the enlarged portion **106** about the longitudinal axis **62** and the transition surface **108** is longitudinally positioned between the inboard entrance **155** and outboard entrance **159**. As such, the fastener **102** and a portion of the enlarged portion **106** is shielded and obscured by the rim **150** leaving only the remainder of the spoke **100** exposed. Thus, the exposed length **170** of the enlarged portion **106** and stepped portion **110** is thereby reduced in comparison with the arrangement of FIGS. **3a-d**. The result is a more aesthetically and visually appealing connection between the spoke **100** and rim **150** as compared to the arrangement of FIGS. **3a-d**. It may be preferred that there remains a longitudinal clearance gap **169** between the transition surface **108** and the bearing face **126** to provide for further threadable adjustment between internally threaded hole **128** and the external threads **104** to adjust spoke tension **58** as needed.

(59) The rim **150** of FIGS. **4a-d** is commonly termed as a “double-wall” rim because it has a spoke bed **152** wall separated from the tire bed **154** wall, with a radial gap or cavity **164** between. It is understood that the present invention may be adapted to rims of other configurations, including those commonly termed as “single-wall” rims where the spoke bed wall and the tire bed wall are shared as a common singular wall.

(60) The embodiment of FIGS. 4e and 4f is identical to the embodiment of FIGS. 4a-d with the exception that spoke hole 296 is stepped to include a laterally enlarged first opening portion 299a and reduced second opening portion 299b, with a laterally projecting transition 307 therebetween. Spoke 100 and nipple 122 are identical to those described in FIGS. 3a-g and 4a-d.

(61) Rim 290 is otherwise schematically identical to rim 150 of FIGS. 4a-d and includes a radially inboard spoke bed 292 wall and a radially outboard tire bed 294 wall and generally radially extending sidewalls to define a generally hollow circumferential cavity 304. Spoke bed 292 is defined by a radially inboard surface 302 and a radially outboard surface 300, including thickness 301 therebetween. It is noted that outboard surface 300 is a curved convex surface as shown here to provide enhanced alignment with the nipple 122 upon assembly. The spoke bed 292 is pierced with a plurality of spoke holes 296 that are stepped to include a longitudinally inboard first opening portion 299a and a longitudinally outboard second opening portion 299b and a laterally extending transition 307 surface therebetween. Second opening portion 299b is sized with diameter dimension 297b to have assembly clearance with the fastener 102. First opening portion 299a is laterally enlarged relative to second opening portion 299b and is sized with diameter dimension 297a for assembly clearance with the enlarged portion 106.

(62) FIG. 4f shows the spoke 100 and nipple 122 assembled to the rim 290 as previously described. Bearing surface 126 is abutting and bearing against outboard surface 300 in an overlie engagement therebetween to support spoke tension 58. Transition surface 108 is longitudinally inboard of the inboard surface 302 such that the enlarged portion 106 is longitudinally overlapping the first opening portion 299a by dimension 305. As such, the fastener 102 and transition surface 108 is obscured by the rim 290 leaving only the remainder of the spoke 100 exposed. Furthermore, the exposed length 308 of the enlarged portion 106 and stepped portion 110 is reduced. The aesthetic and aerodynamic benefit is similar to that described in FIGS. 4a-d. It may be preferred that there remains a longitudinal clearance gap 306 between the transition surface 108 and the transition 307 to provide for further threadable adjustment between internally threaded hole 128 and the external threads 104 to adjust spoke tension 58.

(63) As compared to spoke hole 136 of rim 150, the second opening portion 299b is reduced, which leaves a larger available surface area of outboard surface 300 for bearing interface with the nipple 122. Also, the smaller second opening portion 299b provides for closer lateral alignment of the fastener 102

(64) FIGS. 5a-d describe another arrangement that eliminates the gap 146 (shown in FIG. 3d) to create a more visually appealing wheel assembly 1. Spoke 100 is generally identical to that described in FIGS. 3a-g and rim 150 is identical to that described in FIGS. 4a-d. Nipple 172 is similar to nipple 122 however, nipple 172 includes recess 180 formed directly therein. Spoke nipple 172 includes an internally threaded hole 178 for threadable engagement with external threads 104 and a bearing surface 176 to provide an abutting overlie engagement with the outboard surface 160 on assembly as shown in FIGS. 5b and 5d. Bearing face 176 has a laterally outboard diameter dimension 177 that is larger than the dimension 157 of spoke hole 156 to provide an overlie engagement with the outboard surface 160 of the spoke bed 152. Nipple 172 includes recess 180 therein of diameter dimension 181 and depth dimension 182. Recess 180 extends longitudinally outwardly of bearing face 176 and diameter 181 is sized to receive the enlarged portion 106. Nipple 172 also includes flats 174 for manipulation with a mating wrench (not shown).

(65) As shown in FIGS. 5b and 5d, fastener 102 of spoke 100 has been assembled longitudinally outwardly through spoke hole 156 while the nipple 172 has been passed longitudinally inwardly through access hole 158 and threadably assembled to fastener 102, with internal threaded hole 178 threadably engaged to external threads 104. Bearing surface 176 is abutting and bearing against outboard surface 160 in a structural overlie engagement therebetween to anchor that spoke 100 and support spoke tension 58. FIGS. 5b and 5d show the spoke 100 as advanced longitudinally

outwardly such that the transition surface **108** is passed through the spoke hole **156** to a location longitudinally outboard of the outboard entrance **159** and of the bearing face **126**. Transition surface **108** is also longitudinally overlapping the recess **180** by dimension **187** such that the enlarged portion **106** is longitudinally overlapping both the full length of the spoke hole **156** and a portion of the depth dimension **182** of recess **180** for a total longitudinal overlap dimension **186**. Transition surface **108** and enlarged portion **106** are longitudinally outboard of outboard surface **160** by dimension **187**, which also corresponds to the longitudinal overlap between recess **180** and enlarged portion **106**. It may be preferred that there remains a longitudinal clearance gap **184** between the transition surface **108** and the bottom of recess **180** to allow for further threadable adjustment between internally threaded hole **128** and the external threads **104** to adjust spoke tension **58** as needed.

(66) As shown in FIGS. **5b** and **5d**, the entirety of fastener **102** and a portion of the enlarged portion **106** are shielded and obscured by the rim **150** leaving only the remainder of the spoke **100** exposed. Furthermore, the exposed length **188** of the enlarged portion **106** and stepped portion **110** is reduced. The result is a more aesthetically and visually appealing connection between the spoke **100** and rim **150**, also with enhanced aerodynamics, as compared to the arrangement of FIGS. **3a-d**.

(67) The embodiments of FIGS. **4a-f** and **5a-d** show the spoke nipple as directly connected and engaged to the spoke bed of the mating rim. Alternatively, an intermediate element may be employed between the spoke nipple and the rim to facilitate the connection therebetween as described in FIGS. **6a-d**, **7a-d**, and **8a-d**.

(68) FIGS. **6a-d** describe yet another arrangement that eliminates the gap **146** (of FIG. **3d**) to create a more visually appealing and aerodynamic wheel assembly **1**. Spoke **100** is identical to that described in FIGS. **3a-g**. Rim **190** is schematically identical to rim **130** of FIGS. **3a-d** and includes a radially inboard spoke bed **192** wall and a radially outboard tire bed **194** wall and generally radially extending sidewalls to define a generally hollow circumferential cavity **204**. Spoke bed **192** is defined by a radially inboard surface **202** and a radially outboard surface **200**, including thickness **206** therebetween. It is noted that outboard surface **200** is a flat surface as shown here to support the washer **210** upon assembly. The spoke bed **192** is pierced with a plurality of spoke holes **196** adapted for connection with their respective spokes **100** via spoke nipples **220**. Spoke holes **196** are sized with diameter dimension **197** that is larger than spoke holes **136** (FIGS. **3a-d**) and is sized with the requisite lateral clearance to receive the enlarged portion **106**. Spoke bed thickness **206** roughly corresponds to the longitudinal length of the spoke hole **196**. The tire bed **194** is pierced by access hole **198** that is preferably aligned with spoke hole **196**, to permit the nipple **220** and washer **210** to be assembled as shown in FIGS. **6b** and **6d**.

(69) Nipple **220** and washer **210** are combined to provide a function similar to nipple **172** of FIGS. **5a-d**, however nipple **220** and washer **210** provide a self-aligning swivel interface **215** for improved spoke **100** alignment with the outboard surface **200**. Nipple **220** is similar to nipple **122** (FIGS. **4a-d**) however, nipple **220** includes a bearing surface **226** with spherical convex geometry as shown. Spoke nipple **220** includes an internally threaded hole **226** for threadable engagement with external threads **104** and a bearing surface **226** to provide an abutting overlie engagement with the washer **210** on assembly as shown in FIGS. **6b** and **6d**.

(70) Washer **210** is a generally annular element that includes a longitudinally outwardly projecting outboard face **213**, a longitudinally inwardly projecting bearing face **214** and a hole **212** therethrough of diameter **223**. Outboard face **213** is a spherically concave surface having a spherical radius matched to the spherical convex radius of bearing surface **226**. Bearing face **214** includes a recess **216** or counterbore therein that surrounds hole **212** and is of diameter dimension **219** and depth dimension **217**. Bearing face **214** is of a lateral diameter dimension **218** that is larger than the diameter **197** of spoke hole **196** to provide an overlie engagement with the outboard surface **200** upon assembly. Diameter dimension **219** is sized to be slightly larger than dimension **116a** to permit enlarged portion **106** to enter and overlap the recess **216**.

(71) As shown in FIGS. **6b** and **6d**, fastener **102** has been assembled through spoke hole **196** while the washer **210** and nipple **220** have been passed through access hole **198**, with internally threaded hole **228** threadably assembled to external threads **104** of fastener **104**. Convex bearing face **226** is nested in concave outboard face **213** in a matched spherical swivel interface **215**. Swivel interface **215** provides for pivoting self-alignment between nipple **220** and washer **210** such that nipple **220** may be aligned with longitudinal axis **62**, while bearing surface **214** may be aligned with the outboard surface **200**. Bearing surface **214** is abutting and bearing against outboard surface **200** in an overlaid engagement therebetween to support spoke tension **58**. Washer **210** may be considered as an intermediate connecting element between the spoke nipple **220** and the rim **190**, where nipple **220** is connected to the washer **210** in an overlaid engagement between bearing face **226** and outboard face **213** and washer **210** is connected to the rim in an overlaid engagement between the bearing face **214** and outboard surface **200**.

(72) FIGS. **6b** and **6d** show the spoke **100** as advanced longitudinally outwardly such that the transition surface **108** is longitudinally outboard of the outboard surface **200** and longitudinally overlapping the recess **216** such that the enlarged portion **106** is longitudinally overlapping both the full length of the spoke hole **196** and a portion of the depth dimension **217** of recess **216** for a total longitudinal overlap dimension **211**. It may be preferred that there remains a longitudinal clearance gap **221** between the transition surface **108** and the bottom of recess **216** to provide for further threadable adjustment between internally threaded hole **228** and the external threads **104** to adjust spoke tension **58** in the conventional manner.

(73) As such, the fastener **102** and a portion of the enlarged portion **106** are both obscured by the rim **190** leaving only the remainder of the spoke **100** exposed. Furthermore, the exposed length **222** of the enlarged portion **106** and stepped portion **110** is reduced. The result is a more aesthetically and visually appealing connection between the spoke **100** and rim **190** as compared to the arrangement of FIGS. **3a-d**.

(74) FIGS. **7a-d** describe another arrangement that eliminates the gap **146** to create a more visually appealing wheel assembly **1**. Spoke **100** and nipple **122** are identical to that described in FIGS. **4a-g**. Rim **130'** is identical to that shown in FIGS. **3a-d** with the exception that spoke hole **136'** has a diameter dimension **137'** that is larger than diameter dimension **137** (of rim **130**) and is sized to provide lateral clearance with the dimension **116a** to permit the enlarged portion **106** to pass within spoke hole **136'**. It is noted that the outboard surface **140** is a concave arcuate surface as shown. The access holes **138** permit the nipple **122** and sleeve **230** to be assembled as shown in FIGS. **7b** and **7d**.

(75) Nipple **122** and sleeve **230** are combined to provide a function similar to nipple **172** of FIGS. **5a-d**. Nipple **122** is identical to that described in FIGS. **4a-f**. Sleeve **230** is a generally annular element that includes a radially outwardly projecting outboard face **233**, a radially inwardly projecting bearing face **234** and a hole or opening **232** therethrough. Outboard face **233** is matched to the bearing face **126**. Bearing face **234** is a convex arcuate saddle-shaped surface that is shown to be matched to the concave arcuate surface of outboard surface **140** surrounding spoke hole **136'**. Bearing face **234** is of a lateral diameter dimension **236** that is larger than the diameter **137'** of spoke hole **136'** to provide an overlaid engagement with the outboard surface **140** upon assembly. Opening dimension **238** is sized to be slightly larger than dimension **116a** to permit enlarged portion **106** to enter and overlap the opening **232**.

(76) As shown in FIGS. **7b** and **7d**, fastener **102** of spoke **100** has been assembled through spoke hole **136'** while the sleeve **230** and nipple **122** have been passed through access hole **138**, with internally threaded hole **128** threadably assembled to external threads **104** of fastener **104**. Convex arcuate bearing face **234** is nested in concave arcuate outboard face **140** in a matched arcuate and saddle-shaped interface **246**. It is preferred that bearing face **234** may swivel relative to bearing face **140** such that the nipple **122** may self-align with the longitudinal axis **62** that coincides with bracing angle **248**. Further, the nested engagement between bearing face **234** and outboard

face **140** also provides an anti-rotation engagement therebetween to restrict the rotation of the sleeve **230** relative to the rim **130'** about the longitudinal axis **62**. This may serve to maintain alignment of the sleeve **230** as the nipple **122** is rotated and threadably adjusted.

(77) Bearing surface **234** is abutting and bearing against outboard surface **140** in a structural overlie engagement therebetween to support spoke tension **58**. Simultaneously, bearing surface **126** is abutting and bearing against outboard face **233** in a structural overlie engagement therebetween to support spoke tension **58**. Sleeve **230** may be considered as an intermediate connecting element between the spoke nipple **122** and the rim **130**, where nipple **122** is connected to the sleeve **230** in an overlie engagement between bearing face **126** and outboard face **233** and sleeve **230** is connected to the rim **130'** in an overlie engagement between the bearing face **234** and outboard surface **140**.

(78) FIGS. **7b** and **7d** show the spoke **100** as advanced longitudinally outwardly such that the transition surface **108** is longitudinally outboard of the outboard surface **140** and longitudinally overlapping the opening **232** such that the enlarged portion **106** is longitudinally overlapping both the full length of the spoke hole **136'** and a portion of the opening **232** for a total longitudinal overlap dimension **240**. It may be preferred that there remains a longitudinal clearance gap **242** between the transition surface **108** and the bearing surface **126** to allow for further threadable adjustment between internally threaded hole **128** and the external threads **104** to adjust spoke tension **58** in the conventional manner.

(79) As such, the fastener **102** is obscured by the rim **130'** leaving only the remainder of the spoke **100** exposed. Furthermore, the exposed length **244** of the enlarged portion **106** and stepped portion **110** is reduced. The result is a more aesthetically and visually appealing connection between the spoke **100** and rim **130'** as compared to the arrangement of FIGS. **3a-d**.

(80) FIGS. **8a-d** describe another arrangement that eliminates the gap **146** (of FIGS. **3a-d**) to create a more visually appealing and aerodynamic wheel assembly. Spoke **250** is similar to spoke **100**, except that spoke **250** does not include a stepped portion **110**. Instead, spoke **250** is geometrically simplified to include: a fastener **252** with external threads **254** and having lateral dimension **258**; and an enlarged portion **256** extending longitudinally inwardly therefrom and having lateral dimension **260**. Transition surface **262** extends laterally outwardly by dimension **264** between the sidewall of the fastener **252** to the sidewall of the enlarged portion **254**.

(81) Nipple **122** is identical to that shown in FIGS. **7a-d**. Rim **130''** is identical rim **130'** shown in FIGS. **7a-d** with the exception that spoke hole **136''** has a diameter dimension **137''** that is larger than diameter dimension **137'** and is sized to provide assembly clearance to allow the collar **274** (having external dimension **280**) to longitudinally overlap and extend therein. The access holes **138** permit the nipple **122** and sleeve **266** to be assembled as shown in FIGS. **8b** and **8d**.

(82) Sleeve **266** is similar to sleeve **230** of FIGS. **7a-d** and is a generally annular element that includes a longitudinally outwardly projecting outboard face **268**, a longitudinally inwardly projecting bearing face **270** and a hole or opening **272** therethrough. Unlike sleeve **230**, sleeve **266** includes a cylindrical collar **274** extending longitudinally inwardly from bearing face **270**.

(83) Outboard face **268** is matched to the bearing face **126**. Bearing face **270** is preferably a convex arcuate surface that is shown to be matched to the concave arcuate surface of outboard surface **140** surrounding spoke hole **136''**. Bearing face **270** is of a lateral diameter dimension **276** that is larger than the diameter **137''** of spoke hole **136''** to provide a structural overlie engagement with the outboard surface **140** upon assembly. Opening dimension **278** is sized to be slightly larger than dimension **260** to provide assembly clearance with the enlarged portion **256**, allowing the enlarged portion **256** to enter and longitudinally overlap the opening **272**. Collar **274** has an external dimension **280** that is sized to be inserted in spoke hole **136''** upon assembly.

(84) As shown in FIGS. **8b** and **8d**, spoke **250** has been inserted longitudinally outwardly through spoke hole **136''** while the sleeve **266** and nipple **122** have been passed longitudinally inwardly through access hole **138**, with internally threaded hole **128** threadably assembled to external threads

254 of fastener 252. Convex arcuate bearing face 270 is nested in concave arcuate outboard face 140 in a matched interface 215. Bearing surface 270 is abutting and bearing against outboard surface 140 in a structural overlie engagement therebetween to support spoke tension 58. Simultaneously, bearing surface 126 is abutting and bearing against outboard face 268, also in a structural overlie engagement therebetween to support spoke tension 58. Sleeve 266 may be considered as an intermediate connecting element between the nipple 122 and the rim 130", where nipple 122 is connected to the sleeve 266 in an overlie engagement between bearing face 126 and outboard face 268 and the sleeve 266 is connected to the rim 130" in an overlie engagement between the bearing face 270 and outboard surface 140.

(85) In contrast to sleeve 230 of FIGS. 7a-d, the collar 274 of sleeve 266 is shown to be inserted within spoke hole 136" in a longitudinally overlapping arrangement therebetween. Thus, collar 274 may serve as a lateral barrier between the enlarged portion 256 and the sidewall of the spoke hole 136". Such a barrier may serve provide a lubricious interface with the enlarged portion 256 to prevent galling and/or excessive friction between the enlarged portion 256 and spoke hole 136" during threadable adjustment between the nipple 122 and fastener 252 and the corresponding longitudinal displacement of the spoke 250. The collar 274 may also serve to provide a galvanic barrier between enlarged portion 256 and spoke hole 136" to prevent galvanic corrosion therebetween.

(86) FIGS. 8b and 8d show the spoke 250 as next advanced longitudinally outwardly such that the transition surface 262 is longitudinally outboard of the outboard surface 140 and longitudinally overlapping the opening 272 such that the enlarged portion 256 is longitudinally overlapping both the full length of the spoke hole 136" and a portion of the opening 272 for a total longitudinal overlap dimension 284. It may be preferred that there remains a longitudinal clearance gap 286 between the transition surface 262 and the bearing surface 126 to allow for further threadable adjustment between internally threaded hole 128 and the external threads 254 to adjust spoke tension 58 in the conventional manner.

(87) As such, the fastener 252 is obscured by the rim 130" and sleeve 266, leaving only the remainder of the spoke 250 exposed. The result is a more aesthetically and visually appealing connection between the spoke 250 and rim 130" as compared to the arrangement of FIGS. 3a-d.

(88) FIGS. 9a-b describes an arrangement similar to FIGS. 4a-d, with the exception that the nipple 80 includes a pilot collar 84 to laterally pilot the nipple 80 relative to the spoke hole 156. Spoke 100 and rim 150 are identical to those shown in FIGS. 4a-d. Nipple 80 includes bearing face 82, internally threaded hole 86, and pilot collar 84. Pilot collar 84 is shown as a circular cylindrical collar of diameter 88 that is sized for insertion assembly within hole 156 as shown in FIG. 9a. Bearing surface 82 serves an identical function as bearing surface 126 (FIGS. 4a-d) and has a diameter 87 that is sized to be larger than diameter 157 of spoke hole 156 such that bearing surface 82 provides a structural overlie engagement with the outboard surface 160 to support spoke tension 58.

(89) Pilot collar 84 is shown in FIG. 9b to longitudinally overlap the spoke hole 156 (by dimension 89) such that it serves to laterally pilot and align the nipple 80 to the spoke hole 156. This ensures that the nipple 80 is advantageously centered with the spoke hole 156 and that the bearing surface 82 does not inadvertently laterally shift relative to the entrance 159, thus maintaining an aligned interface between the bearing face 82 and the outboard surface 160. Enlarged portion 106 is longitudinally overlapping the spoke hole 156 by dimension 90.

(90) While my above description contains many specificities, these should not be construed as limitations on the scope of the invention, but rather as exemplifications of embodiments thereof. For example:

(91) The embodiments herein show the spoke hole of the rim to be a circular spoke hole of circular cylindrical profile. This is particularly advantageous because the spoke hole may be formed in a conventional drilling operation. However, the spoke hole may alternatively be noncircular and of

any profile desired. For example, the spoke hole may have a noncircular profile, where the noncircular profile may advantageously be matched to the corresponding cross-section profile of the spoke to minimize clearances therebetween for enhanced aesthetics and aerodynamics. Further, the elongated hole profile may serve to key provide a rotationally keyed engagement to limit relative rotation therebetween about the longitudinal axis.

(92) The embodiments herein show a spoke having an enlarged portion at its second end that is longitudinally overlapping a spoke hole of the rim. It is understood that alternatively or additionally, the hub may include a spoke bed and a spoke hole therethrough. In such case, the spoke may include an enlarged portion at its first end in an arrangement similar to the enlarged portion of the second end shown in FIGS. 3e-g (for example). As such, the enlarged portion may longitudinally overlap the spoke hole of the hub in a transposed arrangement similar to any of the embodiments described herein.

(93) It is to be understood that the invention is not limited to the illustrations described and shown herein, which are deemed to be merely illustrative of the best modes of carrying out the invention, and which are susceptible of modification of form, size, arrangement of parts and details of operation. The invention rather is intended to encompass all such modifications that are within its spirit and scope as defined by the claims.

Claims

1. A vehicle wheel, comprising: a peripheral wheel rim; a central hub with an outer flange; a plurality of spokes extending between said rim and said hub, wherein at least one of said spokes has a first end portion thereof connected to said hub and a second end portion thereof opposed to said first portion and connected to said rim, and a longitudinal axis along said spoke, a span portion between said rim and said hub, and a tensile axis of applied tensile load along said span portion; a bracing element including at least a portion of one of said rim and said hub; wherein said bracing element includes a spoke bed wall having: a longitudinally outboard bed surface; a longitudinally inboard bed surface; and a spoke hole therethrough having a first entrance adjacent said inboard bed surface and a second entrance longitudinally outboard of said first entrance; wherein said spoke includes: a first spoke portion having a first sidewall surface and a first lateral dimension across said first sidewall surface; a second spoke portion longitudinally inboard of said first spoke portion and having a second sidewall surface and a second lateral dimension across said second sidewall surface; wherein said second sidewall surface projects laterally outwardly of said first sidewall surface to include a laterally outwardly projecting transition surface therebetween; wherein said spoke extends through said spoke hole and at least a portion of said transition surface is longitudinally outboard of said first entrance; and wherein said spoke is connected to said bracing element at said first spoke portion.
2. The vehicle wheel according to claim 1, wherein said second spoke portion longitudinally overlaps said spoke hole.
3. The vehicle wheel according to claim 1, wherein said transition surface is entirely located longitudinally outboard of said second entrance.
4. The vehicle wheel according to claim 1, wherein said spoke hole extends along a hole axis and has a circular cylindrical profile about said hole axis.
5. The vehicle wheel according to claim 1, wherein said spoke bed wall is a singular wall between said inboard bed surface and said outboard bed surface.
6. The vehicle wheel according to claim 1, wherein said spoke hole circumscribes said second spoke portion about said longitudinal axis.
7. The vehicle wheel according to claim 1, wherein said rim includes a radially inboard periphery and wherein said radially inboard periphery of said rim includes said inboard bed surface.
8. The vehicle wheel according to claim 1, wherein said spoke hole is a laterally variable opening,

including a enlarged hole portion adjacent said inboard surface that is laterally outward of a reduced hole portion that is longitudinally outward of said enlarged hole portion, wherein said enlarged hole portion is laterally enlarged relative to said reduced hole portion, and wherein said transition surface is longitudinally outboard of said inboard surface and longitudinally overlapping said enlarged hole portion.

9. The vehicle wheel according to claim 1, including a spoke nipple threadably engaged to said first spoke portion, and wherein said spoke nipple includes a bearing surface to provide an overlie engagement with said spoke bed to support said tensile load.

10. The vehicle wheel according to claim 9, wherein said overlie engagement is directly between said bearing surface and said outboard bed surface.

11. The vehicle wheel according to claim 9, wherein said spoke nipple includes a nipple recess that extends longitudinally outboard and laterally inboard of said bearing surface, and wherein said recess is sized to receive at least one of said transition surface and said second spoke portion.

12. The vehicle wheel according to claim 11, wherein at least a portion of said transition surface is longitudinally outboard of said bearing surface and is longitudinally overlapping said spoke nipple within said recess.

13. The vehicle wheel according to claim 9, including an intermediate connecting element, wherein said spoke nipple is engaged to said intermediate connecting element and said intermediate connecting element is engaged to said bracing element to support said tensile load.

14. The vehicle wheel according to claim 13, wherein said intermediate connecting element includes an intermediate bearing surface to provide an overlie engagement with said spoke bed to support said tensile load.

15. The vehicle wheel according to claim 14, wherein said intermediate connecting element includes a connector recess that extends longitudinally outboard and laterally inward of said intermediate bearing surface, and wherein said recess is sized to receive said second spoke portion.

16. The vehicle wheel according to claim 15, wherein at least a portion of said transition surface is longitudinally overlapping said intermediate connecting element within said connector recess.

17. The vehicle wheel according to claim 13, wherein said intermediate connecting element includes a longitudinally extending intermediate opening therethrough, said intermediate opening is sized to receive said second spoke portion.

18. The vehicle wheel according to claim 17, wherein said intermediate connecting element is engaged to said bracing element at a first intermediate bearing interface wherein said intermediate opening is a laterally variable opening, including an enlarged opening portion adjacent said first intermediate bearing interface that is laterally outward of a necked opening portion longitudinally outward of said enlarged opening portion, wherein said transition surface is longitudinally outboard of said intermediate bearing interface and longitudinally overlapping said enlarged opening portion.

19. The vehicle wheel according to claim 13, including a swiveling interface between said nipple and said intermediate connecting element.

20. The vehicle wheel according to claim 13, including a swiveling interface between said intermediate connecting element and said bracing element.

21. The vehicle wheel according to claim 13, wherein said nipple may be rotated independently of said intermediate connecting element about said longitudinal axis.

22. The vehicle wheel according to claim 1, wherein said spoke includes a third spoke portion longitudinally inward of said second spoke portion and having a third sidewall surface and a third lateral dimension across said third sidewall surface, wherein said third sidewall surface is laterally outboard of said second sidewall surface, including a stepped surface therebetween.

23. The vehicle wheel according to claim 13, wherein said intermediate connecting element includes a longitudinally inwardly projecting collar that is laterally inboard of said bearing surface and wherein said collar longitudinally overlaps said spoke hole.

24. The vehicle wheel according to claim 9, wherein said spoke nipple includes a longitudinally

inwardly projecting pilot collar projecting longitudinally inwardly from said bearing surface to longitudinally overlap said spoke hole.

25. The vehicle wheel according to claim 13, wherein intermediate connecting element has an anti-rotation engagement with said rim to restrict rotation of said intermediate connecting element relative to said rim about said longitudinal axis.

26. The vehicle wheel according to claim 1, wherein at least a portion of said second spoke portion is obscured by at least one of said rim and a sleeve connected to said rim.

27. The vehicle wheel according to claim 1, including a connecting element connected to said spoke, wherein said first portion is connected to said connecting element at a first connection interface and said connecting element is connected to said spoke bed at a second connection interface to support said tensile load.
